

vesselFM: A Foundation Model for Universal 3D Blood Vessel Segmentation

Bastian Wittmann¹ Yannick Wattenberg²
 Tamaz Amiranashvili^{1,3} Suprosanna Shit¹ Bjoern Menze¹
¹University of Zurich ²ETH Zurich ³Technical University of Munich
 {bastian.wittmann, bjoern.menze}@uzh.ch

Abstract

Segmenting 3D blood vessels is a critical yet challenging task in medical image analysis. This is due to significant imaging modality-specific variations in artifacts, vascular patterns and scales, signal-to-noise ratios, and background tissues. These variations, along with domain gaps arising from varying imaging protocols, limit the generalization of existing supervised learning-based methods, requiring tedious voxel-level annotations for each dataset separately. While foundation models promise to alleviate this limitation, they typically fail to generalize to the task of blood vessel segmentation, posing a unique, complex problem. In this work, we present vesselFM, a foundation model designed specifically for the broad task of 3D blood vessel segmentation. Unlike previous models, vesselFM can effortlessly generalize to unseen domains. To achieve zero-shot generalization, we train vesselFM on three heterogeneous data sources: a large, curated annotated dataset, data generated by a domain randomization scheme, and data sampled from a flow matching-based generative model. Extensive evaluations show that vesselFM outperforms state-of-the-art medical image segmentation foundation models across four (pre-)clinically relevant imaging modalities in zero-, one-, and few-shot scenarios, therefore providing a universal solution for 3D blood vessel segmentation.

1. Introduction and Motivation

Blood vessel segmentation represents a (pre-)clinically relevant task in (bio)medical image analysis as it plays a vital role in analyzing, diagnosing, and treating various vascular disorders, such as stroke [11], cerebral aneurysms [34], viral pneumonia [36], coronary artery disease [54], and Alzheimer’s [47]. Despite advances in medical image analysis, accurate and robust segmentation of fully-connected vasculature in task-specific imaging modalities still remains a challenging problem, especially in 3D. This is primarily due to the complexity introduced by intricate minuscule vascular geometries, as well as significant domain gaps

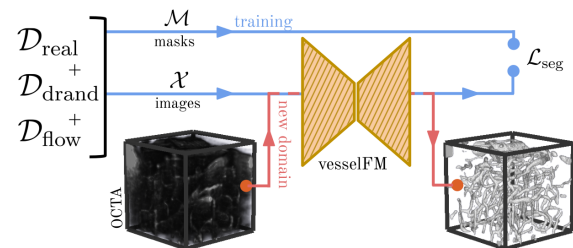


Figure 1. VesselFM is trained in a supervised manner on image-mask pairs from three heterogeneous data sources ($\mathcal{D}_{\text{real}}$, $\mathcal{D}_{\text{drnd}}$, and $\mathcal{D}_{\text{flow}}$) and subsequently applied in a zero-, one-, or few-shot fashion to new, unseen 3D blood vessel domains.

caused by imaging modality and protocol-specific variations in signal-to-noise ratios, vascular patterns and scales, imaging artifacts, and background tissues. These variations typically prevent supervised deep learning-based methods from generalizing to unseen 3D blood vessel domains [51]. Consequently, researchers and clinicians frequently find themselves forced to default to the labor-intensive process of acquiring manual, voxel-level consistent annotations from scratch for analyzing vascular images at hand.

Foundation models for image segmentation pre-trained on large-scale datasets have recently emerged as tools that can effortlessly generalize to unseen data distributions [25]. Although segmentation foundation models also established themselves in the medical field [32, 50, 57], they typically fail to overcome the unique challenges posed by the characteristics of the vascular network. To address this limitation, we propose vesselFM, a Foundation Model precisely tailored to universal 3D blood vessel segmentation. We train vesselFM in a supervised manner on image-mask pairs from three heterogeneous data sources (see Fig. 1). To this end, we first curate $\mathcal{D}_{\text{real}}$, which represents, to the best of our knowledge, the largest annotated dataset for 3D blood vessel segmentation, covering a broad range of imaging modalities from various anatomical regions of different organisms. Second, we supplement $\mathcal{D}_{\text{real}}$ with two synthetic data sources, $\mathcal{D}_{\text{drnd}}$ and $\mathcal{D}_{\text{flow}}$. In particular, we aim to compre-

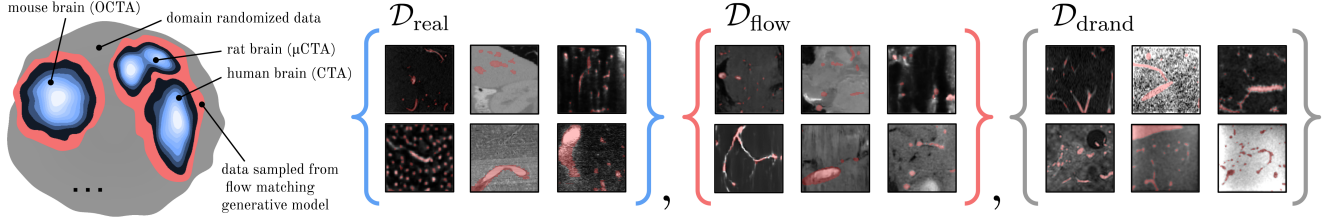


Figure 2. Schematic distributions of our three data sources $\mathcal{D}_{\text{real}}$ (shades of blue), $\mathcal{D}_{\text{flow}}$ (red), and $\mathcal{D}_{\text{drnd}}$ (gray). While we aim to comprehensively cover the general domain of 3D vascular images with $\mathcal{D}_{\text{drnd}}$, $\mathcal{D}_{\text{flow}}$ effectively broadens the distributions of $\mathcal{D}_{\text{real}}$. Note that segmentation masks are shown in translucent red in the exemplary images.

hensively cover the general domain of 3D vascular images by adopting strategies from domain randomization, while we aim to additionally broaden data distributions included in $\mathcal{D}_{\text{real}}$ by sampling from a mask- and class-conditioned flow matching-based generative model (see Fig. 2). Constructing vesselFM on our proposed data sources results in robust features that enable strong generalization to unseen imaging domains, facilitating broad use. In extensive experiments, we demonstrate vesselFM’s state-of-the-art performance on the tasks of *zero*-, *one*-, and *few*-shot blood vessel segmentation across four (pre-)clinically relevant datasets. Our contributions can be summarized as follows:

1. We propose a universal foundation model for 3D blood vessel segmentation capable of *zero*-shot generalization. By open-sourcing checkpoints and code, we aim to provide a foundation model that serves as a robust, out-of-the-box tool for researchers and clinicians alike.¹
2. $\mathcal{D}_{\text{real}}$: We curate the largest dataset for 3D blood vessel segmentation, consisting of carefully processed, real 3D vascular images with matching voxel-level annotations.
3. $\mathcal{D}_{\text{drnd}}$: We propose an elaborate domain randomization strategy tailored to 3D blood vessel segmentation.
4. $\mathcal{D}_{\text{flow}}$: We introduce mask- and class-conditioned flow matching to 3D medical image generation, producing high-fidelity image-mask pairs that adhere to coherent anatomical constraints.

2. Related Works

In this section, we discuss works closely related to vesselFM and elaborate on how vesselFM distinguished itself from existing literature.

2.1. Foundation Models for Image Segmentation

The advent of the Segment Anything Model (SAM) [25] led to the creation of several SAM-like foundation models designed for medical image segmentation [32, 49, 58], even in 3D [48, 50, 53, 57]. The general-purpose segmentation model SAM-Med3D [50], *e.g.*, is trained on a combination of 94 datasets, offers generalization across anatomical

structures and imaging modalities, and claims *zero*-shot transferability to unseen tasks. In contrast to SAM-Med3D, MedSAM-2 [57] relies on the updated SAM 2 [38] and follows the philosophy of treating 3D medical images as videos, resulting in state-of-the-art results while maintaining exceptional generalization across a wide variety of imaging modalities. VISTA3D [22], on the other hand, is developed specifically for CT scans. VISTA3D segments 127 structures and lesions in highly variant CT scans, offering accurate out-of-the-box results and effortless adaptation to unseen structures. In the realm of vessel segmentation, however, foundation models remain under-explored. Earlier works experimented with fine-tuning models pre-trained on vascular data [23, 43], *few*-shot learning [1], and SAM-like methods tailored to 2D OCTA images [49], 3D vEM images [48], and 2D X-ray images [58]. Unlike vesselFM, all above-mentioned vessel segmentation methods are either limited to specific imaging modalities and anatomical structures or have a significantly narrower scope.

2.2. Synthetic Medical Image Generation

In the medical domain, synthetic data is often used to address data scarcity or enhance data diversity. In this context, deep generative models, with diffusion models at the forefront, have emerged as a powerful technique for producing vast amounts of high-fidelity synthetic data [15, 18]. However, to leverage data generated by diffusion models for the task of segmentation, precisely matching image-mask pairs are required. Tackling this challenge, Med-DDPM [13] and SegGuidedDiff [26] integrate semantic conditioning via channel-wise concatenation of the segmentation mask to the model input, resulting in image-mask pairs following coherent anatomical constraints. While Med-DDPM is tailored to 3D brain imaging synthesis, SegGuidedDiff experiments with 2D breast MRI and abdominal CT generation. The concept of domain randomization [44] represents another promising technique in which foreground and background intensity values and morphological features are randomized in a semi-controlled manner to generate versatile synthetic image-mask pairs that can be utilized to train generalist segmentation models resilient to domain shifts. Billot *et al.* [4]

¹<https://github.com/bwittmann/vesselFM>

Table 1. Overview of $\mathcal{D}_{\text{real}}$, including selected dataset statistics. We estimate the mean shape over the x -, y -, and z -axis individually. The voxel size represents the spatial resolution at acquisition time, while the number of patches reflects the approximate amount of 128^3 sub-volumes comprising $\mathcal{D}_{\text{real}}$. We further provide an estimate on label quality focusing on vessel connectivity and annotation precision and a brief overview of dataset-specific pre-processing steps, ensuring that datasets comply with our quality standards. Additional details are provided in Suppl. A. Note that the first four datasets are exclusively used to evaluate vesselFM on unseen domains in our experiments.

	Name	Class c	Tissue Type	Imaging Modality	# Images	Mean Shape	Voxel Size	# Patches	Label Quality	Pre-Processing**
<i>Evaluation</i>	SMILE-UHURA [9]	1	human brain	MRA	14	$640 \times 482 \times 163$	$0.30 \times 0.30 \times 0.30$ mm	335	9	-
	BvEM [48]	2	mouse brain	vEM	1	$3571 \times 5145 \times 2495$	$0.25 \times 0.25 \times 0.32$ μ m	21858	8	r, c
	OCTA [16, 51]	3	mouse brain	OCTA	6	$160 \times 160 \times 160$	$2.00 \times 2.00 \times 2.00$ μ m	11	10	-
	MSD8 [2]	4	human liver	CT	443	$512 \times 512 \times 70$	$0.80 \times 0.80 \times 5.00$ mm	2640	9	r, c, ic
	TubeTK [7]	5	human brain	MRA	42	$896 \times 896 \times 256$	$0.50 \times 0.50 \times 0.80$ mm	4116	8	r, mp
	tUbeNet [23]	6	mouse liver	HREM MRI	1	$400 \times 400 \times 89$	$0.90 \times 0.90 \times 5.00$ mm	6	6	-
		7	mouse brain	two-photon microscopy	1	$500 \times 500 \times 356$	$0.20 \times 0.46 \times 5.20$ μ m	42	7	r, mp
	TopCoW [56]	8	human brain	CTA	90	$334 \times 451 \times 128$	$0.45 \times 0.45 \times 0.13$ mm	863	8	r
		9	human brain	MRA	90	$406 \times 522 \times 128$	$0.30 \times 0.30 \times 0.60$ mm	1179	8	r
	VesSAP [45]	10	mouse brain	light-sheet microscopy (EB)*	19	$500 \times 500 \times 50$	$2.83 \times 2.83 \times 4.99$ μ m	113	7	ic
		11	mouse brain	light-sheet microscopy (WGA)*	19	$500 \times 500 \times 50$	$2.83 \times 2.83 \times 4.99$ μ m	113	7	ic
	DeepVesselNet [43]	12	human brain	MRA	40	$544 \times 514 \times 132$	$0.31 \times 0.31 \times 0.60$ mm	726	7	r
		13	rat brain	μ CTA	4	$256 \times 256 \times 256$	$0.70 \times 0.70 \times 0.70$ mm	32	7	-
	HR-Kidney [27]	14	mouse kidney	X-ray	1	$4608 \times 4608 \times 7168$	$1.60 \times 1.60 \times 1.60$ μ m	72576	6	mp
	3D-IRCADb-01 [41]	15	human liver	CT	20	$512 \times 512 \times 141$	$0.57 \times 0.57 \times 1.60$ mm	352	6	c, mp
	DeepVess [17]	16	mouse brain	multi-photon microscopy	1	$256 \times 256 \times 200$	$1.00 \times 1.00 \times 1.70$ μ m	6	8	-
	CSD [6, 10]	17	human brain	MRA	45	$1024 \times 1024 \times 92$	$0.26 \times 0.26 \times 0.80$ mm	2070	7	-
	VesselExpress [42]	18	mouse brain	light-sheet microscopy	4	$2000 \times 2000 \times 501$	$2.00 \times 2.00 \times 8.00$ μ m	3822	6	-
		19	mouse heart	light-sheet microscopy	3	$250 \times 250 \times 222$	$2.00 \times 2.00 \times 8.00$ μ m	19	6	-
		20	mouse bladder	light-sheet microscopy	10	$300 \times 300 \times 101$	$2.00 \times 2.00 \times 8.00$ μ m	43	6	-
	MiniVess [37]	21	mouse brain	two-photon microscopy	70	$512 \times 512 \times 43$	$0.70 \times 0.70 \times 5.00$ μ m	380	7	-
	HiP-CT [55]	22	human kidney	CT	3	$1350 \times 1311 \times 1844$	$2.50 \times 2.50 \times 2.50$ μ m	4225	8	-
	LS [5]	23	mouse brain	light-sheet microscopy	1	$175 \times 170 \times 200$	$6.00 \times 6.00 \times 6.00$ μ m	2	9	ic

* different dyes used for staining: wheat germ agglutinin (WGA) and Evans blue (EB); ** r: resampled, c: cropped, mp: mask post-processed (e.g., smoothed or multi-class labels to binary), ic: intensities clipped

were the first to adopt a domain randomization strategy for medical image segmentation by proposing SynthSeg, a model capable of segmenting brain MRI scans of varied resolution and contrast. AnyStar [12] extends SynthSeg’s concept to 3D instance segmentation of star-convex shapes such as nuclei, nodules, or metastases. In this work, we leverage both deep generative models and domain randomization strategies to enrich our data sources. Specifically, we extend Med-DDPM by introducing the concept of flow matching [29, 30], which has shown to be superior to diffusion [14, 33], to anatomically controllable vascular image generation, and adapt the concept of domain randomization for 3D blood vessel segmentation.

3. Data Source Generation

We train vesselFM on three heterogeneous data sources: 1) diverse real data ($\mathcal{D}_{\text{real}}$), 2) domain randomized data ($\mathcal{D}_{\text{drand}}$), and 3) data sampled from a flow matching-based generative model ($\mathcal{D}_{\text{flow}}$). Below, we detail each of these three data sources.

3.1. $\mathcal{D}_{\text{real}}$: Diverse Real Data

The development process of generalist foundation models necessitates large-scale, diverse real datasets [8, 32, 50]. To this end, we curate $\mathcal{D}_{\text{real}} = \{\mathcal{X}_{\text{real}}, \mathcal{M}_{\text{real}}\}$, encompassing real images $\mathcal{X}_{\text{real}}$ and their corresponding annotated segmentation masks $\mathcal{M}_{\text{real}}$ (see Table 1 for dataset overview and statistics; see Fig. 3 for exemplary images). $\mathcal{D}_{\text{real}}$ comprises more than 115,000 3D patches of shape 128^3 curated from 17 annotated sources, which we further separate into 23 datasets based on tissue types, imaging modalities, and

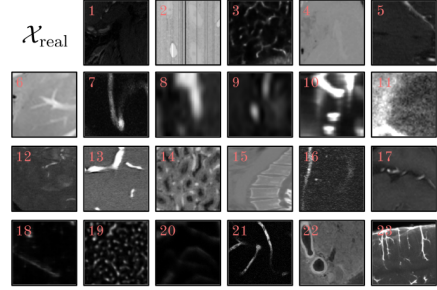


Figure 3. Slices of images $\mathcal{X}_{\text{real}}$ from $\mathcal{D}_{\text{real}}$. $\mathcal{D}_{\text{real}}$ contains vascular images of shape 128^3 with matching voxel-level annotations collected from 23 datasets (classes are indicated in red) of diverse imaging modalities, depicting a wide range of anatomical regions.

protocols. For ease of reference, each dataset in $\mathcal{D}_{\text{real}}$ is indexed by a unique class $c \in \mathcal{C} = \{1, \dots, 23\}$ (see Table 1, 2nd column). Importantly, $\mathcal{D}_{\text{real}}$ covers a broad array of clinically (e.g., MRA, CTA, and X-ray) and pre-clinically relevant (e.g., vEM, μ CTA, and two-photon microscopy) imaging modalities. It integrates data from several anatomical regions (e.g., brain, kidney, and liver) in various organisms (e.g., human, mouse, and rat), thus providing an expansive spectrum of blood vessel patterns of varying structural and functional properties. Further, we deliberately include datasets of the same imaging modalities to bridge domain gaps in, e.g., scale and contrast, induced by high variability in dataset-specific imaging protocols. We pay special attention to solely including datasets that adhere to a high standard in label quality. To curate $\mathcal{D}_{\text{real}}$, we pre-process each dataset and finally extract patches of our target shape (128^3)

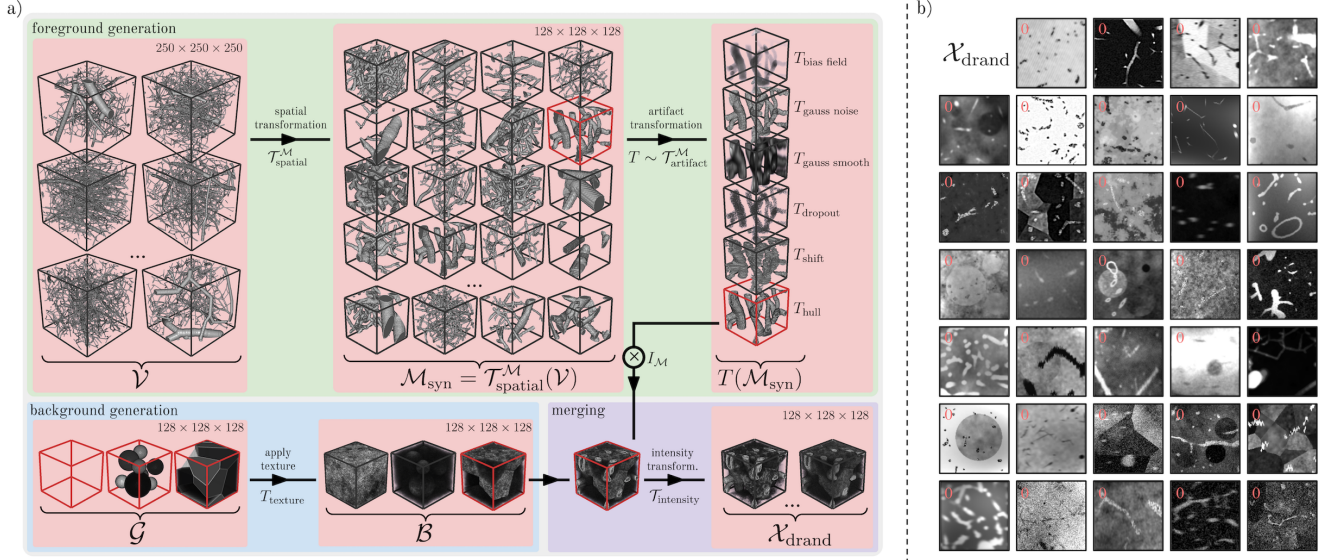


Figure 4. a) Schematic overview of our domain randomized generative pipeline used to generate $\mathcal{D}_{\text{drand}} = \{\mathcal{X}_{\text{drand}}, \mathcal{M}_{\text{syn}}\}$. We specifically highlight its three main components: foreground generation, background generation, and merging. Note that we indicate instances forwarded to the subsequent step in the color red for illustration purposes. b) Slices of exemplary images $\mathcal{X}_{\text{drand}}$, categorized as $c = 0$. The wide variety of generated, highly diverse images showcases the effectiveness of our proposed domain randomization strategy.

from the images and their corresponding labels. Details on pre-processing can be found in Suppl. A. To the best of our knowledge, $\mathcal{D}_{\text{real}}$ represents the largest real dataset for the task of 3D blood vessel segmentation to this date.

3.2. $\mathcal{D}_{\text{drand}}$: Domain Randomization

Inspired by recent works [4, 12], we explore the use of domain randomization to generate a massive amount of matching image-mask pairs of semi-randomized style, categorizing them under class $c = 0$. In the following, we describe our proposed domain randomization strategy tailored to 3D blood vessels. An overview of our domain randomized generative pipeline used to create $\mathcal{D}_{\text{drand}} = \{\mathcal{X}_{\text{drand}}, \mathcal{M}_{\text{syn}}\}$ is depicted in Fig. 4. We detail its parametrization in Suppl. B.

Foreground generation. To generate synthetic masks acting as foreground geometries, we utilize 1,137 vascular patches $\mathcal{V}$ of shape 250^3 provided by Wittmann *et al.* [51]. These vascular patches, originating from graph representations of corrosion casts [46], accurately preserve both general angioarchitectural and morphological properties characteristic of 3D blood vessels with minimal artifacts. Therefore, $\mathcal{V}$ provides the perfect foundation by ensuring functional fidelity, a key requirement for generating realistic vascular images. First, we process $\mathcal{V}$ by applying spatial transformations $\mathcal{T}_{\text{spatial}}^{\mathcal{M}}$. Specifically, we crop with random center positions to the target shape of 128^3 , followed by random flipping and rotation across all three axes. To ensure robustness against variations in blood vessel scale and

density, we subsequently apply random dilation and random zooming. Additionally, we address variations in blood vessel curvature and tortuosity by employing random elastic deformation and binary smoothing. Applying $\mathcal{T}_{\text{spatial}}^{\mathcal{M}}$ results in a broad range of realistic vascular patterns (see Fig. 4a). Throughout this work, we refer to $\mathcal{T}_{\text{spatial}}^{\mathcal{M}}(\mathcal{V})$ as the set $\mathcal{M}_{\text{syn}}$. Next, we emulate a broad range of foreground artifacts present in real vascular images by concluding with carefully selected artifact transformations $\mathcal{T}_{\text{artifact}}^{\mathcal{M}} = \{T_{\text{bias field}}, T_{\text{gauss noise}}, T_{\text{gauss smooth}}, T_{\text{dropout}}, T_{\text{shift}}, T_{\text{hull}}, T_{\text{identity}}\}$ (see Fig. 4a). In contrast to $\mathcal{T}_{\text{spatial}}^{\mathcal{M}}$, which is applied consecutively, we sample a single artifact transformation T from $\mathcal{T}_{\text{artifact}}^{\mathcal{M}}$ for each processed vascular patch.

Background generation. Considering that the interplay of imaging techniques and protocols, background tissue compositions, and pathological conditions creates a broad spectrum of background intensity patterns, we model background images $\mathcal{B}$ containing various background geometries of diverse textures. Specifically, we incorporate three variants of background geometries $\mathcal{G}$ (see Fig. 4a). 1) Spheres: we include non-overlapping spheres; 2) polyhedrons: we split the image into polyhedral regions using Voronoi partitioning [3]; 3) none: we do not incorporate any background geometries. For background geometries and the background itself, we sample versatile Perlin noise patterns [12, 35] that accurately mimic textures characteristic of vascular images. Further, we also include plain background images consisting of a randomly selected intensity

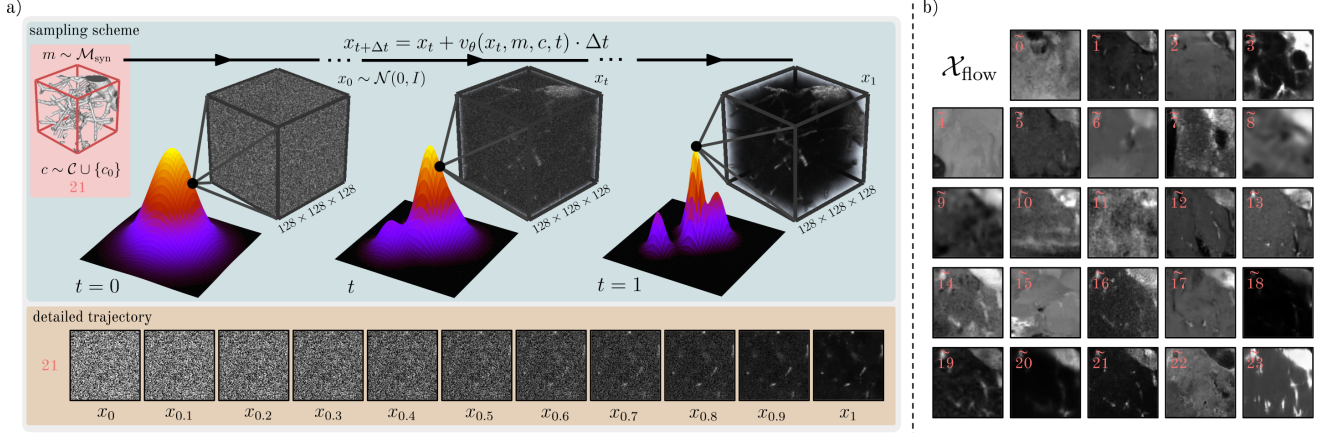


Figure 5. a) Sampling of synthetic images $\mathcal{X}_{\text{flow}}$ via our mask- and class-conditioned flow matching-based generative model. We explicitly show our sampling scheme, mapping a sample $x_0 \sim \mathcal{N}(0, I)$ to an exemplary sample x_1 of class 21. In addition, we present a more detailed trajectory, which is for improved visibility plotted in 2D. b) Slices of exemplary images $\mathcal{X}_{\text{flow}}$, sampled from our generative model. Note that all of the depicted slices are conditioned on the same mask, and we solely vary the class. We would like to emphasize that our generative model is able to produce synthetic images almost indistinguishable from real images (compare with Fig. 3).

value drawn from $\mathcal{U}(0, 1)$ for enhanced diversity.

Fore- and background merging. Subsequently, we merge $T(\mathcal{M}_{\text{syn}})$ into the background images $\mathcal{B}$ via either voxel-wise addition/subtraction or by replacing background intensity values with mask intensity values. To assign intensities $I_{\mathcal{M}}$ to individual masks that separate them from their respectively matched background images, we estimate mean background intensities $I_{\mathcal{B}}^\mu$ and follow $I_{\mathcal{M}} \notin [I_{\mathcal{B}}^\mu - \delta, I_{\mathcal{B}}^\mu + \delta]$. Lastly, we additionally aim to broaden the domain of the merged images by applying an ample range of intensity transformations $\mathcal{T}_{\text{intensity}}$ with loose configurations. We consecutively perform random bias field augmentations, add Gaussian noise, apply random localized spikes in k-space, randomly adjust the image contrast, perform Gaussian smoothing with either individual or shared σ values for all spatial dimensions, add Rician noise, apply Gibbs noise, perform random Gaussian sharpening, and randomly transform intensity histograms.

3.3. $\mathcal{D}_{\text{flow}}$: Flow Matching-Based Image Generation

Flow matching [29, 30] is a promising alternative to diffusion models and has shown superior performance on natural images [14, 33]. To generate our third data source $\mathcal{D}_{\text{flow}} = \{\mathcal{X}_{\text{flow}}, \mathcal{M}_{\text{syn}}\}$, we train and subsequently sample images from a mask- and class-conditioned flow matching-based generative model $\mathcal{F}$, aiming at further broadening the distributions of $\mathcal{D}_{\text{real}}$ in a data-driven manner. $\mathcal{F}$ utilizes a θ -parametrized network representing a learned, time-dependent velocity field v , which is trained to map samples $x_0 \sim \mathcal{N}(0, I)$ to samples x_1 of the data distribution via an

ordinary differential equation (ODE):

$$\frac{d}{dt}x_t = v_\theta(x_t, m, c, t), \quad (1)$$

where $t \in [0, 1]$ represents the time, c the class we condition on, and m the conditioning mask. To train $\mathcal{F}$, we optimize the conditional flow matching (CFM) objective [29], which minimizes the L_2 loss between the predicted velocity $v_\theta(x_t, m, c, t)$ and the sampled ground truth velocity $u_t(x_t|x_1)$ at time t :

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, x_1, x_t} \|v_\theta(x_t, m, c, t) - u_t(x_t|x_1)\|^2. \quad (2)$$

We define the forward process as $x_t = tx_1 + (1-t)x_0$, leading to $u_t(x_t|x_1) = (x_1 - x_t)/(1-t)$ in the loss above. The time-linear forward process provides straighter ODE trajectories than a popular variance-preserving diffusion noise schedule in DDPM, simplifying sampling at inference [29, 30, 33]. We train $\mathcal{F}$ on matching image-mask pairs (x_1, m) and their associated classes $c \in \mathcal{C} \cup \{0\}$, sampled from *both* of our previously generated data sources. Building on anatomically controllable medical image generation methods [13, 26], we implement mask conditioning by concatenating the mask channel-wise with the input image x_t . Class information is incorporated by adding the class embedding to the time embedding, followed by injection into the intermediate feature layers via addition.

To generate $\mathcal{D}_{\text{flow}}$, we ultimately sample a vast amount of images $\mathcal{X}_{\text{flow}}$ (see Fig. 5) by discretizing (1) via Euler integration:

$$x_{t+\Delta t} = x_t + v_\theta(x_t, m, c, t) \cdot \Delta t, \quad \Delta t = \frac{1}{N}, \quad (3)$$

Table 2. Quantitative results. We compare vesselFM to state-of-the-art foundation models for medical image segmentation on three tasks: *zero*-, *one*-, and *few*-shot 3D blood vessel segmentation. VesselFM is evaluated on four datasets of clinical (SMILE-UHURA [9], MSD8 [2]) and pre-clinical (OCTA [16, 51], BvEM [48]) relevance and consistently outperforms all baselines by a relatively large margin.

Task	Model	OCTA [16, 51]		BvEM [48]		SMILE-UHURA [9]		MSD8 [2]	
		Dice $\uparrow$	clDice $\uparrow$	Dice $\uparrow$	clDice $\uparrow$	Dice $\uparrow$	clDice $\uparrow$	Dice $\uparrow$	clDice $\uparrow$
<i>zero</i> -shot	tUbeNet [23]	36.01	23.64	10.03	11.17	48.32	36.85	5.13	5.84
	VISTA3D [22]	13.60	3.72	0.94	2.03	5.05	1.62	23.83	20.25
	SAM-Med3D [50]	6.74	6.56	5.98	7.38	2.12	1.66	7.94	7.89
	MedSAM-2 [57]	28.56	15.76	10.92	12.27	3.85	5.46	14.53	14.27
	vesselFM (ours)	46.94	67.07	67.49	62.04	74.66	75.27	29.69	36.14
<i>one</i> -shot	tUbeNet [23]	38.09	59.37	10.75	11.53	57.67	53.25	13.66	15.41
	VISTA3D [22]	51.24	25.69	8.25	15.04	56.53	42.42	31.73	32.94
	SAM-Med3D [50]	38.33	54.90	49.47	52.14	38.57	36.94	29.29	36.78
	MedSAM-2 [57]	56.68	50.95	24.07	24.69	19.78	11.87	30.21	23.89
	vesselFM (from scratch)*	65.57	73.79	63.85	39.55	37.99	45.72	27.13	29.48
	vesselFM (ours)	72.10	83.73	78.27	79.91	76.43	78.36	36.88	48.65
<i>few</i> -shot	tUbeNet [23]	41.61	57.98	5.41	10.22	56.31	49.28	17.67	18.97
	VISTA3D [22]	54.25	32.59	24.04	38.10	61.17	51.05	41.90	46.45
	SAM-Med3D [50]	37.85	56.94	57.86	66.04	46.59	44.63	31.30	35.48
	MedSAM-2 [57]	56.96	51.99	18.76	19.66	58.15	42.72	29.24	22.38
	vesselFM (from scratch)*	67.37	75.79	63.03	56.69	50.51	58.77	32.64	36.03
	vesselFM (ours)	75.70	84.03	78.11	84.54	78.77	79.37	45.04	57.25

*Model not pre-trained on $\mathcal{D}_{\text{real}}$, $\mathcal{D}_{\text{drand}}$, and $\mathcal{D}_{\text{flow}}$.

where N represents the total number of time steps. Given that $\mathcal{M}_{\text{syn}}$ covers the required range of blood vessel patterns and is devoid of annotator-induced biases and errors in segmentation masks, we opt to exclusively use masks $m \sim \mathcal{M}_{\text{syn}}$ during sampling. For clarity, we use tilde to denote classes of data generated by $\mathcal{F}$ (e.g., $\tilde{7}$).

4. Experiments and Results

In this section, we elaborate on our findings and showcase vesselFM’s performance on three tasks: *zero*-shot, *one*-shot, and *few*-shot segmentation. We evaluate vesselFM on four 3D blood vessel segmentation datasets of unseen clinically (SMILE-UHURA [9], MSD8 [2]) and pre-clinically (OCTA [16, 51], BvEM [48]) relevant domains. In this context, we extract three patches of shape 128^3 from each of these evaluation datasets and use the remaining data for testing and validation (see Suppl. A for details). With the three extracted patches, we define the *one*- and *few*-shot segmentation task as fine-tuning models on either one or all three patches. For *zero*-shot evaluation, we apply models out-of-the-box on the test data without prior fine-tuning. This setup mimics clinical scenarios where annotated data is scarce.

We compare vesselFM to four foundation models designed for 3D medical image segmentation: the generalizable 3D blood vessel segmentation model tUbeNet [23], the CT-specific VISTA3D [22], and the two SAM-like general-purpose segmentation models SAM-Med3D [50] and MedSAM-2 [57]. In our experiments, we exclude the classes of the four datasets used for evaluation (see Table 1, upper section) from the curation of $\mathcal{D}_{\text{real}}$ and $\mathcal{D}_{\text{flow}}$. We generate $\mathcal{D}_{\text{flow}}$ by sampling 10,000 image-mask pairs from $\mathcal{F}$ on

a single RTX A6000 GPU over the course of three days. To curate $\mathcal{D}_{\text{drand}}$, we sample 500,000 image-mask pairs from our domain randomized generative pipeline. All images-mask pairs are of shape 128^3 . We train vesselFM using all three data sources, with weights assigned roughly according to their sizes (70% $\mathcal{D}_{\text{drand}}$, 20% $\mathcal{D}_{\text{real}}$, and 10% $\mathcal{D}_{\text{flow}}$). We opt for MONAI’s re-implementation of the UNet architecture proposed by Isensee *et al.* [24] to present our segmentation model. For flow matching, we use the UNet from MedDDPM [13] to represent the learned velocity field v . We set the total number of time steps N in (3) to 100. Following common practices, we report Dice scores and topology-aware centerline Dice (clDice) scores [40], which assess the preservation of tubular appearance and connectivity of blood vessels. Further details on our experimental setup can be found in Suppl. C.

4.1. Quantitative and Qualitative Results

Quantitative results can be observed in Table 2. We find that our proposed foundation model, vesselFM, tailored to universal 3D blood vessel segmentation, outperforms the baseline models on all datasets and tasks by a large margin.

Zero-shot task. VesselFM exhibits exceptional *zero*-shot generalization on all four datasets, which cover a diverse array of unseen domains and even imaging modalities (OCTA and vEM). Surprisingly, vesselFM scores 5.86 Dice points higher than VISTA3D on MSD8, even though VISTA3D was trained on 11,454 CT volumes, including data from MSD8 itself. This highlights vesselFM’s strong inductive bias, enabled by training on our three proposed hetero-

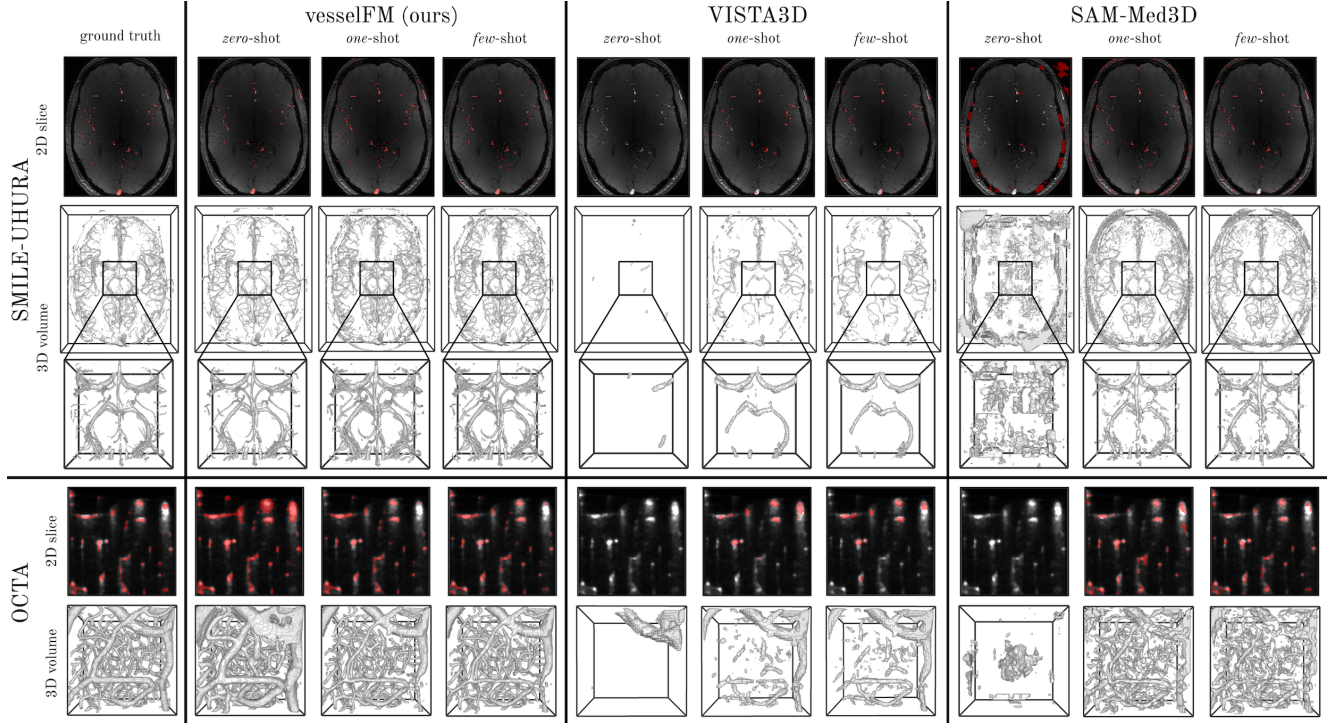


Figure 6. Qualitative results (better viewed zoomed in). We visualize predictions on the SMILE-UHURA and the OCTA datasets for all three tasks (*zero*-, *one*-, and *few*-shot). We compare vesselFM to VISTA3D, SAM-Med3D, and ground truth segmentation masks. Note that vesselFM consistently predicts state-of-the-art results, even in the *zero*-shot setting, demonstrating exceptional generalization to unseen domains. For better visibility, we show a zoomed-in view of the 3D predictions on the SMILE-UHURA datasets.

geneous blood vessel data sources. Moreover, vesselFM outperforms the generalizable 3D blood vessel segmentation model tUbeNet, trained on four blood vessel datasets of varying imaging modalities. We observe that tUbeNet struggles in more complex imaging modalities where blood vessels do not have a stark contrast to background tissues (e.g., BvEM and MSD8). The general-purpose segmentation models SAM-Med3D and MedSAM-2 both fail to segment blood vessels in the *zero*-shot setting. Notably, vesselFM’s *zero*-shot results surpass *few*-shot results achieved by baseline models on SMILE-UHURA in Dice and cIDice.

One- and few-shot tasks. Fine-tuning vesselFM in a *one*- or *few*-shot manner generally increases segmentation performance. Given that some baseline models rely on networks, which may easily overfit to the small amount of training data provided in the *one*- and *few*-shot settings, we additionally compare vesselFM to a variant of the same configuration that is trained from scratch without being pre-trained on our three proposed data sources (see footnote in Table 2; see Suppl. D for more insights). We observe that neglecting our three proposed data sources causes a notable decrease in Dice and cIDice scores, validating our rationale.

Qualitative results mirror the insights gained from

quantitative results. Specifically, we find that vesselFM demonstrates exceptional *zero*-shot generalizations, free of annotator-specific biases (see Fig. 6).

4.2. Ablation Studies

We ablate vesselFM’s design choices on the SMILE-UHURA [9] dataset, targeting brain vessel segmentation in MRA images, a task of high clinical importance for automated diagnosis of various diseases, such as aneurysms. All ablations are conducted for the *zero*-shot segmentation task, given that *zero*-shot generalization is the most pivotal component of segmentation foundation models.

First, we ablate the relevance of our three proposed data sources, $\mathcal{D}_{\text{real}}$, $\mathcal{D}_{\text{drand}}$, and $\mathcal{D}_{\text{flow}}$ (see Table 3). To this end, we train vesselFM on each data source individually and progressively augment $\mathcal{D}_{\text{real}}$ with $\mathcal{D}_{\text{drand}}$ and $\mathcal{D}_{\text{flow}}$. A more complete ablation of our data sources covering all four evaluation datasets and all tasks can be found in Suppl. E. We observe that supplementing $\mathcal{D}_{\text{real}}$ with $\mathcal{D}_{\text{drand}}$ and $\mathcal{D}_{\text{flow}}$ results in an impressive increase in Dice and cIDice scores of 9.21 and 11.74, respectively. This highlights that leveraging all our three proposed heterogeneous data sources collaboratively allows vesselFM to learn robust features that are well-suited for *zero*-shot generalization, enabling a founda-

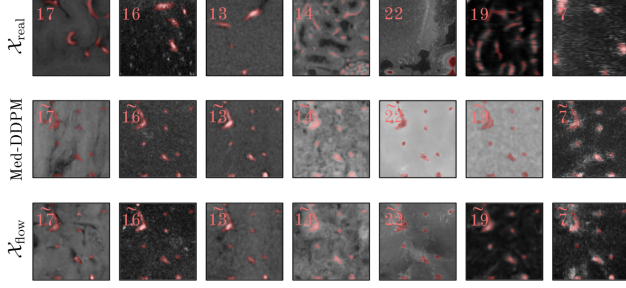


Figure 7. Qualitative comparison of images generated by our flow matching-based generative model $\mathcal{F}$ ($\mathcal{X}_{\text{flow}}$, 3rd row) with images generated by the diffusion-based generative model Med-DDPM [13] (2nd row). We also include real images of the same classes for reference ($\mathcal{X}_{\text{real}}$, 1st row). For improved comparability, we consistently condition $\mathcal{F}$ and Med-DDPM on the same mask m . Segmentation masks are displayed in translucent red.

Table 3. Ablation of data sources.

Data sources	Dice $\uparrow$	clDice $\uparrow$
$\mathcal{D}_{\text{real}}$	65.45	63.53
$\mathcal{D}_{\text{real}} + \mathcal{D}_{\text{drand}}$	69.38	72.10
$\mathcal{D}_{\text{real}} + \mathcal{D}_{\text{drand}} + \mathcal{D}_{\text{flow}}$	74.66	75.27
$\mathcal{D}_{\text{drand}}$	55.34	63.16
$\mathcal{D}_{\text{flow}}$	14.33	16.14

tion model for universal 3D blood vessel segmentation.

Second, we ablate the design choices of our proposed mask- and class-conditioned flow matching-based generative model $\mathcal{F}$, used to sample $\mathcal{D}_{\text{flow}}$ (see Table 4). Specifically, we exclude $\mathcal{D}_{\text{drand}}$ from training of $\mathcal{F}$ (2nd row), utilize real masks $\mathcal{M}_{\text{real}}$ instead of our proposed synthetic masks $\mathcal{M}_{\text{syn}}$ during sampling from $\mathcal{F}$ (3rd row), and exclude class conditioning (4th row). Subsequently, we replace $\mathcal{D}_{\text{flow}}$ with the respective generated variants and re-train vesselFM. Omitting $\mathcal{D}_{\text{drand}}$, and consequently the class

Table 4. Ablation of mask- and class-conditioned flow matching.

Method used for $\mathcal{D}_{\text{flow}}$	Dice $\uparrow$	clDice $\uparrow$
$\mathcal{F}$	74.66	75.27
$\mathcal{F}$, no $\mathcal{D}_{\text{drand}}$	71.24	73.93
$\mathcal{F}$, $\mathcal{M}_{\text{real}}$	70.12	70.84
$\mathcal{F}$, no class cond.	74.56	74.75
Med-DDPM [13]	70.34	73.35

$c = 0$, from training reduces the Dice score by 3.42. This suggests that the added data diversity and quantity provided by $\mathcal{D}_{\text{drand}}$ enables $\mathcal{F}$ to generate a wider variety of images, thereby facilitating generalization. Generating $\mathcal{D}_{\text{flow}}$ by conditioning on real masks $\mathcal{M}_{\text{real}}$ included in $\mathcal{D}_{\text{real}}$ instead of synthetic masks results in a 4.54 reduction in Dice score. This finding supports our decision to utilize synthetic masks sampled from $\mathcal{M}_{\text{syn}}$, which are, unlike masks from

$\mathcal{M}_{\text{real}}$, free of annotator-induced errors and offer greater diversity. We find that omitting class conditioning leads to a modest drop in Dice by 0.10. Finally, we compare $\mathcal{F}$, relying on the concept of flow matching, to the diffusion-based, generative model Med-DDPM [13] (5th row). $\mathcal{F}$ outperforms Med-DDPM not only quantitatively (4.32 Dice) but also qualitatively (see Fig. 7). Specifically, we observe that Med-DDPM frequently struggles to capture class-specific artifacts accurately, resulting in low-fidelity synthetic images (e.g., see Fig. 7, classes 22 and 19).

Third, we ablate vesselFM’s segmentation model by experimenting with relevant medical 3D segmentation networks (see Table 5). Our employed UNet variant surpasses

Table 5. Ablation of vesselFM’s segmentation model.

Segmentation arch.	Dice $\uparrow$	clDice $\uparrow$
UNet	74.66	75.27
SwinUNETR [19]	60.00	53.92
SwinUNETR-V2 [21]	74.54	74.80
UNETR [20]	46.74	40.15
3D UX-Net [28]	49.99	46.31
MedNeXt [39]	56.47	61.95

transformer-based [19–21] and ConvNeXt [31]-based [28, 39] networks alike, which accurately represents the current landscape in medical image segmentation.

5. Conclusion and Outlook

In this work, we propose vesselFM, a foundational model for universal 3D blood vessel segmentation. VesselFM is capable of accurately segmenting 3D vasculature in previously unseen domains and performs superior to state-of-the-art medical image segmentation foundational models. We enable *zero-shot* generalization by training vesselFM on three proposed heterogeneous data sources ($\mathcal{D}_{\text{real}}$, $\mathcal{D}_{\text{drand}}$, and $\mathcal{D}_{\text{flow}}$), which we extensively ablate in our experiments. Given that vesselFM pushes the frontier in the (pre-)clinically relevant task of 3D blood vessel segmentation, we hope that our work enables novel insights into vascular disorders and fosters the development of advanced diagnostic tools, ultimately resulting in improved patient outcomes (see Suppl. F for discussion on clinical utility). We advise future work to experiment with tailored post-processing steps to improve blood vessel connectivity (potentially at the graph level [52]) and extend vesselFM to multi-class or instance segmentation tasks.

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